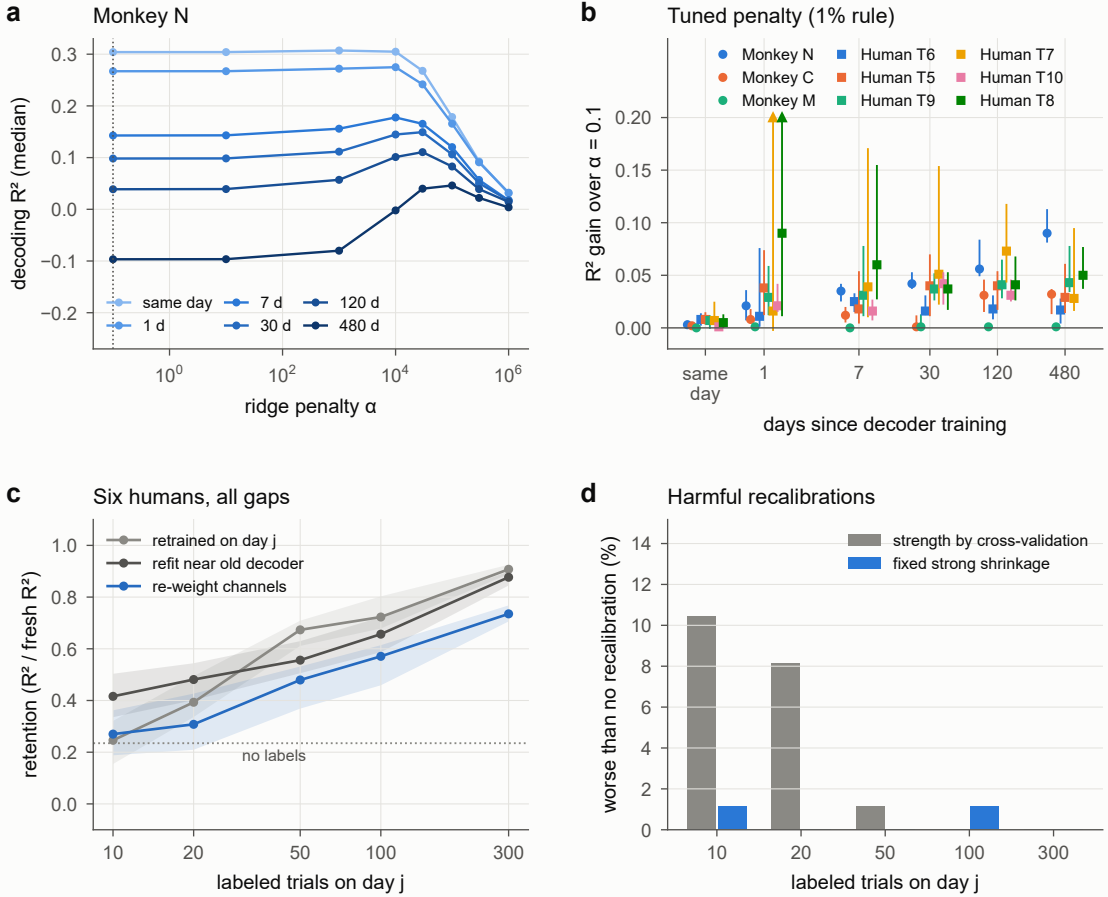

a
Monkey N
decoding R² (median)
0.3
0.2
0.1
0.0
−0.1
−0.2
same day
1 d
7 d
30 d
120 d
480 d
10⁰
10²
10⁴
10⁶
ridge penalty α
b
Tuned penalty (1% rule)
Monkey N
Monkey C
Monkey M
Human T6
Human T5
Human T9
Human T7
Human T10
Human T8
R² gain over α = 0.1
0.20
0.15
0.10
0.05
0.00
same day
1
7
30
120
480
days since decoder training
c
Six humans, all gaps
retrained on day j
refit near old decoder
re-weight channels
retention (R² / fresh R²)
1.0
0.8
0.6
0.4
0.2
0.0
no labels
10
20
50
100
300
labeled trials on day j
d
Harmful recalibrations
strength by cross-validation
fixed strong shrinkage
worse than no recalibration (%)
14
12
10
8
6
4
2
0
10
20
50
100
300
labeled trials on day j